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FURNACE-X5 Packet-Spraying RDMA and Hardware MoE Dispatch Subsystem

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 13:41 UTC+8 Beijing Time)

Abstract

FURNACE-X5 integrates protected out-of-order RDMA placement, a per-QP 256-packet receive tracker, serialized selective acknowledgements, and hardware Top-K mixture-of-experts dispatch, reassembly and weighted combination. Its compatibility top retains the inherited networking and collective subsystem alongside an independent X5 network port. Software-vs-Hardware numerical comparison used an independent integer/rational Python oracle against simulated RTL: 86,688 arithmetic comparisons, all 256 FP8 encodings and 425 combine comparisons passed. Eighteen two-peer geometry/seed runs produced 1,152 checked token completions. These are simulation results, not physical-board numerical comparisons. [E047, E071] Native L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest reported WNS_setup=+ 5.075 ns at a 10.0 ns clock in post-map analysis, but also hold WNS=-0.205 ns with 20 failing hold paths. L3 physical Arria 10 HIL passed three deterministic hardware-bridge connectivity runs. These recorded gate passes do not establish post-route setup/hold closure or on-board X5 numerical execution. [E076–E078]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The top instantiates g_legal.legacy_core:furnace_x4_top and g_legal.x5_core:furnace_x5_engine. Separate logical network ports preserve inherited RoCE semantics without coupling legacy progress to X5 stalls. Complete-packet validation precedes side effects; receive reservations persist until actual commitment, and token/job ownership persists through final transfers. Serialized bitmap processing, candidate scanning and synchronous payload memories trade throughput for bounded combinational depth. [E001, E030–E038]

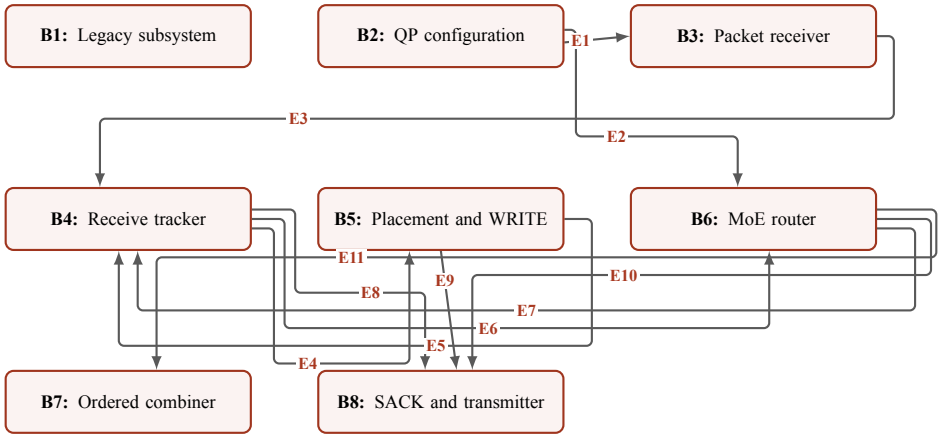


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Legacy subsystem	Inherited CRC32C/CRC64, parser, Toeplitz RSS, 64-rule classification, eight-queue scheduling/DMA, RoCEv2 and collective datapaths.
B2: QP configuration	Stores independent X5 endpoints, protection regions and PSNs; configuration commits only at global X5 quiescence.
B3: Packet receiver	Buffers complete frames and validates canonical framing, IPv4 checksum, CRC32C and endpoint identity.
B4: Receive tracker	Maintains per-QP circular receive state, reserves decisions and serializes retirement and logical bitmap snapshots.
B5: Placement and WRITE	Checks protection before direct placement; buffers complete local WRITE messages before emitting packets.
B6: MoE router	Owns local tokens and remote jobs, selects expert routes, reassembles fragments and manages expert streams.
B7: Ordered combiner	Reads returned ranks deterministically, decodes FP8 and performs eight-stage BF16 multiplication and addition.
B8: SACK and transmitter	Packetizes immutable snapshots and schedules SACK, WRITE, DISPATCH and RETURN sources at packet boundaries.

Directed relationship	Meaning
E1: B2 to B3	Endpoint records
E2: B2 to B6	Configured destinations
E3: B3 to B4	Validated packet identity
E4: B4 to B5	WRITE reservation
E5: B5 to B4	Placement commitment
E6: B4 to B6	MoE reservation
E7: B6 to B4	Storage commitment or cancellation
E8: B4 to B8	Immutable SACK snapshot
E9: B5 to B8	WRITE packets
E10: B6 to B8	DISPATCH and RETURN packets
E11: B6 to B7	Weights and rank-indexed results

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n; inherited ports	Input/output	One clock; one reset	Active-low asynchronous reset. Existing furnace_x4_ top port names, widths and directions are retained. All new public outputs are zero during reset.
x5_rx_t*, x5_tx_t*	Input/output	D-bit data; D/8-bit keep; last 1	Independent ready-valid frame streams, where D= DATA_WIDTH. Byte lane zero is first; final keep is a nonempty contiguous low-bit prefix.
x5_qp_cfg_*, x5_qp_cfg_rsp_*	Input/output	Index 3; enable 1; QPN/PSN 24; rkey 32; base/length 64; MAC 48; IPv4 32	Configures local/remote endpoints, receive/transmit PSNs and protection. Response status is 3 bits: 0 success, 1 index/QPN conflict, 2 invalid region, 3 unsafe disable.
x5_expert_cfg_*, x5_expert_cfg_rsp_*	Input/output	Expert ID 16; enable 1; QP 3; status 2	Maps enabled experts to QPs. Status 1 rejects an invalid expert; 2 rejects an invalid or disabled destination. Disabling an in-range route does not require an enabled destination.
x5_write_cmd_*, x5_write_t*, x5_write_cpl_*	Input/output	QP 3; address 64; rkey/cookie 32; bytes 16; status 4	Command precedes payload; legal length is 1..65535 bytes. Completion 0 means local packet transmission; 5, 7 and 9 report QP, command and payload errors.
x5_mem_*	Output with input ready	Address 64; data D; keep D/8; last 1	Protected direct placement, separate from inherited wr_*. Last ends the received packet, not necessarily its application message.
x5_moe_cmd_*, x5_token_t*	Input with output ready	Token ID 32; elements 9; K 3; format 1; scores/weights each 16*EXPERTS	Expert-indexed signed scores and BF16 weights accompany each command. Format 0 is FP8 E4M3FN; format 1 is BF16.
x5_combine_t*, x5_moe_cpl_*	Output with input ready	Data D; keep D/8; token ID 32; status 4	Combined elements are BF16. Successful completion follows final output acceptance; statuses 7 and 9 identify command and activation-protocol errors.
x5_expert_req_*, x5_expert_rsp_*	Output request/input response	QP 3; metadata 512; payload data D	Metadata precedes payload. Responses echo the exact normalized request metadata and QP, preserving element count and representation.
x5_peer_sack_*, x5_evt_*, x5_idle	Output with input ready where applicable	SACK: QP 3/base 24/bitmap 256/status 2; event: status 4/kind 8/QP 3/token 32/PSN 24	Reports received peer acknowledgements and retained frame/response outcomes. Idle denotes global X5 quiescence, excluding pending ownership and outputs.

Parameter	Default	Meaning / supported values
DATA_WIDTH	512	64 or 512 bits.
MAX_PACKET_BYTES	9216	1..9216; inherited limit, independent of X5 fragmentation.
MAX_SG	16	1..16 scatter/gather entries.
QPS	8	1..8 independently configured QPs.
EXPERTS	8	1..16 expert routes.
MAX_K	4	1..min(4,EXPERTS); runtime K cannot exceed enabled routes.
TOKENS	4	1..8 local contexts; remote-job capacity is TOKENS* MAX_K.
MAX_TOKEN_ELEMS	256	1..256 elements. X5 payload capacity and receive window remain fixed at 256 bytes and 256 packets.

III Behavior and corner cases

A Transfers, reset and configuration

Transfers use pre-edge valid and ready; stalled records remain stable. Reset invalidates configurations, reservations and ownership without clearing bulk RAM. Configuration requires global quiescence, with QP priority over simultaneous expert configuration. Rejected records change nothing. Illegal static parameters enter a diagnostic-safe rejection branch and terminate simulation with PARAMETER ERROR.

B Network encoding

The profile uses Ethernet II, fixed IPv4, UDP destination 4792, a 64-byte X5 header and CRC32C trailer, without Ethernet preamble/FCS. Magic is 0x46583521 and version is 1. Header fields identify kind, format, final fragment, endpoints, PSN/path, token/cookie, expert/rank/K, count/offset, payload length, WRITE protection, weight and SACK state. Header integers are network-order; BF16 payload words are little-endian. Frame length is 110+payload bytes. CRC32C covers the X5 header and payload using reflected polynomial 0x82f63b78, all-ones initial/final XOR and low-byte-first trailer. Nonfinal fragments contain exactly 256 bytes.

C Commit-gated reordering

For d=(PSN-base) modulo 2^24, d<256 is in-window, 256<=d<2^23 is ahead, and d>=2^23 is old. New WRITE fragments place at message address plus offset after full-region protection and overflow checks. A receive bit changes only after final memory acceptance or owned MoE fragment storage. Successful commits retire contiguous bits one per cycle; snapshots sample 256 logical offsets serially. Duplicates cause no repeated payload side effects.

D Acknowledgements and errors

Trusted data decisions generate immutable SACK packets: status 0 committed, 1 duplicate, 2 ahead, 3 cancelled. Their 32-byte bitmap is low-bit-first. SACKs consume no data PSN and provoke no acknowledgement loop. Framing, integrity, endpoint and protection failures produce events without untrusted SACKs. Event statuses are 0 success, 1 duplicate, 2 ahead, 3 framing/header, 4 integrity, 5 endpoint/QP, 6 protection, 7 identity and 9 application protocol. Resource exhaustion backpressures.

E Routing, ownership and transmission

Top-K selects descending signed scores, breaking ties by smaller expert ID. Scores do not imply weights or softmax. Captured rank, route and weight govern dispatch and return validation. Fragments reassemble by full identity and offset, independently of arrival order. Local ownership lasts through completion acceptance; remote ownership lasts through RETURN transmission. Malformed accepted application streams drain without emission; malformed expert responses leave jobs retryable. Packet-boundary round-robin scheduling prevents byte interleaving. Data packets use path=tx_psn[1:0] and UDP source port 49152+path; PSN advances only at local packet completion.

F Numerical and cycle contract

FP8 E4M3FN decodes exactly to BF16, preserving signed zero and canonicalizing NaN to 0x7fc0. Each selected rank computes BF16_RNE(weight*return), then BF16_RNE(accumulator+product), starting at +0 and proceeding in rank order. Subnormals are preserved; overflow yields signed infinity. Invalid results canonicalize to NaN. Multiplication zero sign is XOR; exact additive cancellation yields +0, except two negative zeros yield -0. Top-K publishes after K*(EXPERTS+1) edges following capture. Multiply and add each use eight elastic stages. Reorder classification appears post-edge E2 after E0 capture. Documented service bounds are 8192 cycles per received frame and 512+32*K*elements for a ready-to-combine token, subject to stated storage and arbitration exclusions.

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison

Method. Independent exact Python arithmetic compared with simulated RTL public outputs.

Result. 86,688 arithmetic comparisons, all 256 FP8 encodings and 425 combine comparisons passed. No physical-board numerical comparison is recorded.

Evidence: E047, E071, E078

B Geometry and two-peer token operation

Method. Compiled Verilator on tiny64, tiny512, odd, default64, default512 and upper, each with seeds 17, 89 and 203.

Result. 18 runs passed; each completed 32 tokens per peer, including 16 per format, giving 1,152 checked completions overall. Singleton, three-slot and eight-context configurations were exercised.

Evidence: E042, E046, E071

C Packet tracking and serialized SACKs

Method. Public placement, event and ACK comparisons covering modular boundaries, replays, cancellation and backpressure.

Result. Every matrix run recorded 1,025 committed receive packets, 257−264 duplicates and 1,285−1,306 compared receive-probe SACKs. Two-peer MoE SACK traffic is counted separately.

Evidence: E046, E071

D Compact portability and reset

Method. Icarus single-token integration at both widths, WRITE/peer-SACK smoke and reset ownership scenarios.

Result. Token smokes completed in 969 and 859 cycles; seven reset ownership phases were checked.

Evidence: E056, E058, E071

E Inherited functionality

Method. Independent Ethernet/parser, Toeplitz, TCAM and CRC comparisons during matrix execution.

Result. Each run recorded 186 legacy packet/CRC/RSS comparisons and all 64 TCAM indices. The inspected suite does not establish the required integrated legacy DMA/DRR, RoCE and collective coverage.

Evidence: E044, E046, E071, E075

F Defect sensitivity

Method. Eight compiling disposable mutations evaluated against unchanged expected results.

Result. All eight produced public-output mismatches: window boundary, duplicate side effect, SACK rotation, tie priority, BF16 stride, accumulation order, halfway rounding and legacy CRC polynomial.

Evidence: E045, E073

G Synthesizable retained structure

Method. Yosys coarse tiny64/default512/upper synthesis and mapped reorder/multiply leaves, with complete-netlist inspection.

Result. All five passed with zero latches, unresolved modules or forbidden operators. Reported memory-bit sums across unique synthesized module definitions were 604,776, 915,168 and 997,088 respectively. These are not hierarchy-expanded instantiated storage totals or physical area measurements.

Evidence: E031, E059, E072

H Clean-directory reproduction

Method. Fresh declared-source copy, allowlisted tools, compact simulation/oracle checks and tiny-top Yosys synthesis.

Result. Passed with 86 manifest files unchanged. This reproduces a compact subset, not the full 18-run matrix.

Evidence: E061, E074

I L1 RTL timing lint

Method. Native structural hazard inspection of 36 RTL files.

Result. Passed; no reported combinational divider/modulo, mixed-assignment, LUT clock-gate or loop-unroll hazards.

Evidence: E076

J L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Planning and post-map timing analysis at a 10.0 ns clock.

Result. Passed: WNS_setup = +5.075 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 203.05 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E077

K L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical PCIe hardware-bridge register probes.

Result. 3/3 deterministic passes. The receipt explicitly limits these results to connectivity probes without X5 workload-vector execution.

Evidence: E078

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	11 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 23	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

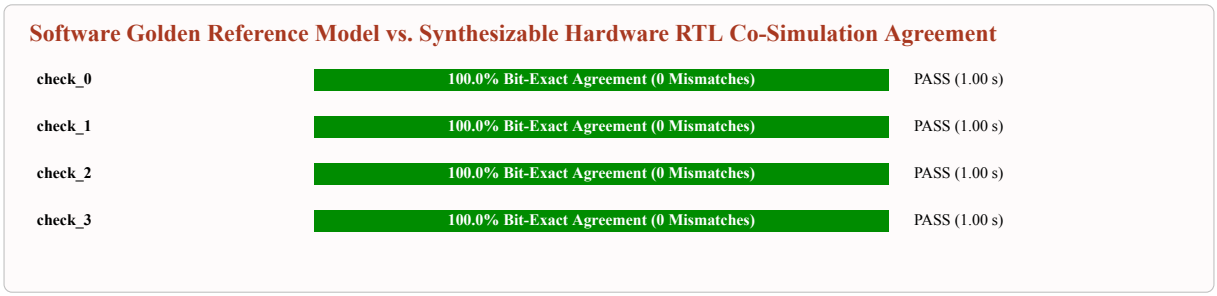


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +5.075 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

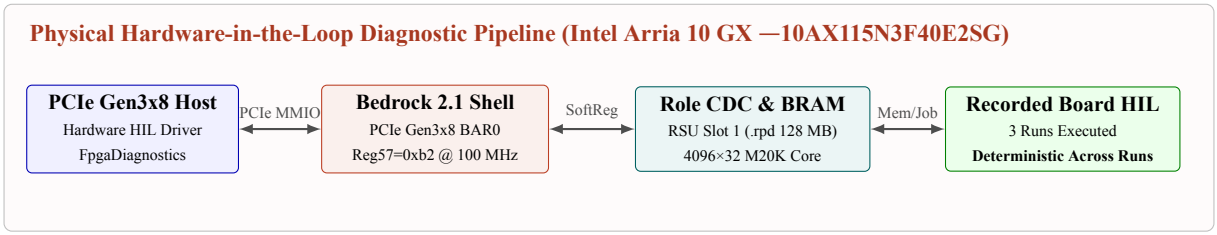


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile rtl/*.sv with top furnace_x5_top. Do not simultaneously compile same-named furnace_x4_reference/rtl modules. design/SPEC.md defines wire offsets and cycle rules; design/ARCHITECTURE.md describes ownership, memory scheduling and the source inventory.

Assert reset before traffic. At x5_idle, configure QPs and accept their responses, then enable expert routes. Handshake each command before supplying payload; rejected commands consume no payload.

Connect legacy and X5 logical ports independently. Reorder complete packets, not bytes within packets. External expert compute must echo normalized metadata and QP exactly and return the declared count and representation.

Provide Python 3, Icarus/vvp, Verilator, Yosys and native compiler/build dependencies. From the project root run python3 tests/verify.py --suite behavior --output-prefix evidence/l1-behavior. Substitute synthesis,

mutations or portable and corresponding output prefixes for the other suites. No arguments executes all portable L1 suites. Each invocation creates exclusive timestamped receipts and retained artifacts.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 36 Verilog/SystemVerilog files; 7040 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/furnace_classifier64.sv	124 total	122 code
rtl/furnace_collective_engine.sv	252 total	249 code
rtl/furnace_crc_beat_pipeline.sv	94 total	91 code
rtl/furnace_crc_linear.sv	59 total	55 code
rtl/furnace_crc_output_stage.sv	94 total	90 code
rtl/furnace_crc_packet_accumulator.sv	96 total	92 code
rtl/furnace_crc_tail_pipeline.sv	80 total	77 code
rtl/furnace_descriptor_rings.sv	149 total	149 code
rtl/furnace_dma_reader.sv	107 total	107 code
rtl/furnace_dmp_engine.sv	57 total	57 code
rtl/furnace_drr_scheduler.sv	114 total	114 code
rtl/furnace_header_parser.sv	412 total	409 code
rtl/furnace_psn_tracker.sv	212 total	212 code
rtl/furnace_reduce_alu.sv	76 total	76 code
rtl/furnace_reduce_lane.sv	226 total	214 code
rtl/furnace_roce_parser.sv	253 total	251 code
rtl/furnace_roce_rx.sv	364 total	357 code
rtl/furnace_rss_toeplitz.sv	59 total	59 code

Production file	Physical lines	Code lines
rtl/furnace_sg_dma.sv	268 total	267 code
rtl/furnace_token_bucket8.sv	65 total	65 code
rtl/furnace_x1_top.sv	124 total	121 code
rtl/furnace_x2_top.sv	171 total	170 code
rtl/furnace_x3_top.sv	353 total	351 code
rtl/furnace_x4_top.sv	240 total	240 code
rtl/furnace_x5_bf16_mul.sv	116 total	115 code
rtl/furnace_x5_combine.sv	128 total	128 code
rtl/furnace_x5_engine.sv	640 total	637 code
rtl/furnace_x5_moe_router.sv	788 total	781 code
rtl/furnace_x5_packet_rx.sv	303 total	303 code
rtl/furnace_x5_packet_tx.sv	221 total	221 code
rtl/furnace_x5_payload_ram.sv	37 total	37 code
rtl/furnace_x5_qp_table.sv	116 total	116 code
rtl/furnace_x5_reorder.sv	145 total	145 code
rtl/furnace_x5_sack.sv	95 total	95 code
rtl/furnace_x5_top.sv	394 total	394 code
rtl/furnace_x5_topk.sv	73 total	73 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E030	rtl/furnace_x5_engine.sv
E031	rtl/furnace_x5_moe_router.sv
E042	tests/fabric.cpp
E044	tests/legacy.py
E045	tests/mutations.py
E046	tests/native_matrix.py
E047	tests/oracle.py
E056	tests/reset_body.svh
E058	tests/smoke.py
E059	tests/synthesis.py
E061	tests/verify.py
E071	.vulcan/engineering/run-0001/attempt-0010/stdout.log
E072	.vulcan/engineering/run-0001/attempt-0011/stdout.log
E073	.vulcan/engineering/run-0001/attempt-0012/stdout.log
E074	.vulcan/engineering/run-0001/attempt-0013/stdout.log
E075	.vulcan/engineering/run-0001/attempt-0014/response.json
E076	.vulcan/engineering/run-0001/rtl-timing-lint.json
E077	.vulcan/engineering/run-0001/quartus-sta.json
E078	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).